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UE21MA251 LINEAR ALGEBRA AND ITS APPLICATIONS

.Unit 4: Orthogonalization , Eigen Values and Eigen Vectors Orthogonal Bases, The Gram- Schmidt Orthogonalization, Spectral theorem, Introduction to Eigenvalues and Eigenvectors, Properties of Eigenvalues and Eigenvectors, Symmetric Matrices, Diagonalization of a Matrix.

Class No.	Portions to be covered
46-48	Orthogonal Bases- Orthogonal Matrices, Properties, Rectangular Matrices with orthonormal columns
49-50	The Gram- Schmidt Orthogonalization, A = QR Factorization.
51-53	Introduction to Eigen values and Eigenvectors, Properties of eigenvalues andEigen vectors, Spectral theorem, Symmetric Matrices, Cayley-Hamilton theorem(Statement only)
54-55	Diagonalization of a Matrix, Powers and Products of Matrices
56	**Applications**
57-58	Matlab class number 7, 8 -Projection least squares, The Gram- Schmidt process
59	Class room Assignment
60	**ISA 4**

Classwork problems:

1.	Verify that $\{(-2,1,3,0),(0,-3,1,-6),(-2,-4,0,2)\}$ is an orthogonal set of vectors in R^4. and use it to construct an orthonormal set of vectors in R^4. Answer: $\frac{1}{\sqrt{14}}v_1, \frac{1}{\sqrt{46}}v_2, \frac{1}{2\sqrt{6}}v_3$
2.	Let W = { (a, b, b) / a , b are real } and let v = (3, 2, 6). (i) Find an orthonormal basis for W (i) Find the projection of v onto W , say v1 (ii) Decompose v into a sum of two vectors v1+ v2 where v2 is projection of v onto $W^\perp$. Answer : v = (3, 4, 4) + (0, -2 , 2)
3. ✓	Find a third column so that the matrix $Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -- \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & -- \\ 0 & \frac{2}{\sqrt{6}} & -- \end{pmatrix}$ is orthogonal. Answer : $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

4.	If W is a subspace spanned by the orthogonal vectors (2, 5, -1) and (-2 , 1, 1) find the point in W that is closest to (1, 2, 3). Answer : (-2/5, 2, 1/5)
5.	What multiple of $a_1 = \begin{pmatrix}1\\1\end{pmatrix}$ should be subtracted from $a_2 = \begin{pmatrix}4\\0\end{pmatrix}$ to make the result orthogonal to a_1? Factor A =QR with orthonormal vectors in Q. Answer : 1 , A=QR=$\begin{pmatrix}\sqrt{2} & 2\sqrt{2}\\0 & 2\sqrt{2}\end{pmatrix}$
6.	Find an orthonormal set q1, q2, q3 for which q1 and q2 span the column space of $A = \begin{pmatrix}1 & 1\\2 & -1\\-2 & 4\end{pmatrix}$ Which fundamental subspace contains q3 ? What is the least squares solution of Ax = b if b = (1, 2, 7) ? Answer : $q_3 = \left(-\frac{2}{3},\frac{2}{3},\frac{1}{3}\right)$. x = (1,2)
7. ✓	Use the Gram – Schmidt process to find a set of orthonormal vectors from the independent vectors $a1 = (0,3,4), a2 = (1,0,1)$ *and* $a3 = (1,1,3)$. Also find A = QR factorization where A = [a1 a2 a3] Answer : $\left(0,\frac{3}{5},\frac{4}{5}\right),\left(\frac{5\sqrt{34}}{\sqrt{34}},\frac{6\sqrt{34}}{85},\frac{9\sqrt{34}}{\sqrt{170}}\right),\left(\frac{-3\sqrt{34}}{\sqrt{34}},\frac{-2\sqrt{34}}{17},\frac{3\sqrt{34}}{34}\right)$
8.	Use Gram-Schmidt Orthogonalization process to transfer the basis $\left(\begin{pmatrix}1 & 0\\0 & 0\end{pmatrix},\begin{pmatrix}0 & 1\\0 & 0\end{pmatrix},\begin{pmatrix}0 & 0\\1 & 0\end{pmatrix},\begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}\right)$ of M_{2*2} into an orthogonal basis if $A^TB = tr(AB^T)$
9.	Find the eigenvalues and the corresponding eigenvectors of $\begin{pmatrix}2 & 0 & 1\\0 & 2 & 0\\1 & 0 & 2\end{pmatrix}$, $\begin{pmatrix}1 & -1 & 1\\1 & 0 & 0\\-1 & 1 & -1\end{pmatrix}$ Answer : Eigen values are (i) 1, 2, 3, Eigen vector(−1,0,1), (0,1,0), (1,0,1) (ii)Eigen values 1, i, −i, eigen vectors, (0,1,1)(−1, i ,1), (−1, −i, 1)
10.	Find the eigenvalues and eigenvectors of $A = \begin{pmatrix}1 & 4\\1 & 1\end{pmatrix}$. Verify that the trace equals the sum of eigenvalues and the determinant equals their product. If we shift A to A − 7 I what are the eigenvalues and eigenvectors and how are they related to those of A ? Also find the eigenvalues and eigenvectors of A^2 , A^{-1} Answer: -1,3, Eigen vector(-2,1),(2,1), Eigen value (-8,-4), Eigen vectors remains same, (1,9),(-1,1/3) eigen vectors remains same.

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11.	Find the characteristic equation and hence find the inverse of $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ using the Cayley – Hamilton's theorem. Answer: $-(\lambda-4)(\lambda+1)(\lambda-1)$, $\begin{pmatrix} -\frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}$
12.	Find the matrix A whose eigen values are 2 and 5, and whose eigen vectors are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ respectively using $S\Delta S^{-1}$. Answer: $A = \begin{pmatrix} 2 & 3 \\ 0 & 5 \end{pmatrix}$
13.	Factor $A = \begin{pmatrix} 3 & 4 \\ -1 & 8 \end{pmatrix}$ into $S\Delta S^{-1}$ and hence compute A^8. Answer: Eigen values are 4,7 and Eigen vectors are $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
14.	Find all eigenvalues and eigenvectors of A = $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ and write two different diagonalizing matrices S. Answer : eigenvalues are 0, 0, 3 with eigenvectors (-1 , 1, 0) , (-1, 0, 1), (1, 1, 1)
15.	Find the matrices S and S^{-1} to diagonalize $A = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$ What are limits of Δ^k and $S\Delta^k S^{-1}$ as $k \to \infty$? Answer : eigenvalues of A are 1 and 0.2 with eigenvectors (1 , 1) , (1 , -1). And $\Delta^k \to \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $S\Delta^k S^{-1} \to \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$

Linear Algebra & Its Applications

Unit 4 - Orthogonalization, Eigen Values & Eigen Vectors

Class work problems

3) Find a third column so that the matrix $Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -- \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & -- \\ 0 & \frac{2}{\sqrt{6}} & -- \end{pmatrix}$ is orthogonal.

Solution: By definition of orthogonal matrix, first & second columns are orthogonal to the required column. Let this third column be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

$$Q = \begin{array}{c} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & x \\ 1/\sqrt{2} & -1/\sqrt{6} & y \\ 0 & 2/\sqrt{6} & z \end{bmatrix} \\ \begin{matrix} a \quad & \quad b \quad & \quad c \end{matrix} \end{array}$$

$$a^T c = 0$$

$$\Rightarrow \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\boxed{x + y = 0}$$

$$b^T c = 0 \Rightarrow \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\Rightarrow x - y + 2z = 0$$

$$x + y + 0 \cdot z = 0$$
$$x - y + 2z = 0$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 2 \end{bmatrix}$$

$$-2y + 2z = 0$$
$$-2y = -2z$$
$$\boxed{y = z}$$

$$x + y = 0$$
$$x = -y$$
$$\boxed{x = -z}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -z \\ z \\ z \end{pmatrix} = z \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

(2) for this column to be orthonormal,

$$(-z)^2+(z)^2+(z)^2=1$$

$$z^2+z^2+z^2=1$$

$$3z^2=1$$

$$z^2=1/3 \Rightarrow z=1/\sqrt{3}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

7) use the Gram-Schmidt process to find a set of orthonormal vectors from the independent vectors $a_1=(0,3,4)$ $a_2=(1,0,1)$ and $a_3=(1,1,3)$. Also find $A=QR$ factorization where $A=[a_1\, a_2\, a_3]$

Solution:

$$q_1 = \frac{a_1}{\|a_1\|} = \frac{1}{\sqrt{9+16}} = \frac{1}{5}\begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix}$$

$$q_2 = \frac{e_2}{\|e_2\|} ; \quad e_2 = a_2 - (q_1^T a_2) q_1$$

$$e_2 = a_2 - (q_1^T \cdot a_2) q_1 ; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{bmatrix} 0 & \frac{3}{5} & \frac{4}{5} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - 4/5 \begin{pmatrix} 0 \\ 3/5 \\ 4/5 \end{pmatrix} \Rightarrow e_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 12/25 \\ 16/25 \end{pmatrix}$$

$$e_2 = \begin{bmatrix} 1 \\ -12/25 \\ 9/25 \end{bmatrix} ; \quad \|e_2\| = \sqrt{\frac{1}{1} + \frac{144}{625} + \frac{81}{625}}$$

$$\|e_2\| = \sqrt{\frac{625+144+81}{625}} = \frac{\sqrt{850}}{25} = \frac{5\sqrt{34}}{25}$$

$$\therefore q_2 = \frac{1}{\sqrt{34}/5}\begin{bmatrix} 1 \\ -12/25 \\ 9/25 \end{bmatrix} = \begin{bmatrix} 5/\sqrt{34} \\ -12/25 \times \frac{5}{\sqrt{34}} \\ 9/25 \times \frac{5}{\sqrt{34}} \end{bmatrix} = \begin{bmatrix} 5/\sqrt{34} \\ -12/5\sqrt{34} \\ 9/5\sqrt{34} \end{bmatrix}$$

$$q_3 = \frac{e_3}{\|e_3\|} ; \quad a_3 - (q_1^T a_3) q_1 - (q_2^T a_3) q_2 = e_3$$

$$e_3 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{bmatrix} 0 & 3/5 & 4/5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix} - \begin{bmatrix} \frac{5}{\sqrt{34}} & \frac{-12}{5\sqrt{34}} & \frac{9}{5\sqrt{34}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \begin{bmatrix} \frac{+5}{\sqrt{34}} \\ \frac{-12}{5\sqrt{34}} \\ \frac{9}{5\sqrt{34}} \end{bmatrix}$$

$$q_3 ; \; e_3 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - 3 \begin{bmatrix} 0 \\ 3/5 \\ 4/5 \end{bmatrix} - \begin{bmatrix} \frac{5}{\sqrt{34}} - \frac{12}{5\sqrt{34}} + \frac{27}{5\sqrt{34}} \end{bmatrix} \begin{bmatrix} \frac{+5}{\sqrt{34}} \\ \frac{-12}{5\sqrt{34}} \\ \frac{9}{5\sqrt{34}} \end{bmatrix}$$

$$e_3 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 9/5 \\ 12/5 \end{pmatrix} - \begin{bmatrix} \frac{25 - 12 + 27}{5\sqrt{34}} \end{bmatrix} \begin{matrix} \frac{+5}{\sqrt{34}} \\ \frac{-12}{5\sqrt{34}} \\ \frac{9}{5\sqrt{34}} \end{matrix}$$

$$e_3 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 9/5 \\ 12/5 \end{pmatrix} - \frac{8}{\sqrt{34}} \begin{bmatrix} 5/\sqrt{34} \\ -12/5\sqrt{34} \\ \frac{9}{5\sqrt{34}} \end{bmatrix}$$

$$e_3 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 9/5 \\ 12/5 \end{pmatrix} - \begin{pmatrix} 40/34 \\ -96/34\times5 \\ 72/170 \end{pmatrix}$$

$$e_3 = \begin{pmatrix} \frac{1}{1} - 0 - 40/34 \\ \frac{1}{1} - 9/5 + 96/170 \\ \frac{3}{1} - \frac{12}{5} - \frac{72}{170} \end{pmatrix} = \begin{pmatrix} \frac{34-40}{34} \\ \frac{170-306+96}{170} \\ \frac{510-408-72}{170} \end{pmatrix} = \begin{bmatrix} \frac{-6}{34} \\ \frac{-40}{170} \\ \frac{30}{170} \end{bmatrix}$$

(4)

$$e_3 = \begin{bmatrix} -3/17 \\ -4/17 \\ 3/17 \end{bmatrix} ; \quad \|e_3\| = \sqrt{\frac{9}{289} + \frac{16}{289} + \frac{9}{289}}$$

$$= \sqrt{\frac{34}{289}} = \frac{\sqrt{2 \times 17}}{17}$$

$$= \frac{\sqrt{2}\sqrt{17}}{\sqrt{17} \times \sqrt{17}} = \sqrt{\frac{2}{17}}$$

$$\therefore \; q_3 = \frac{1}{\sqrt{2}/\sqrt{17}} \begin{bmatrix} -3/17 \\ -4/17 \\ 3/17 \end{bmatrix} = \begin{bmatrix} \frac{-3 \times \sqrt{17}}{\sqrt{2} \times 17} \\ \frac{-4 \times \sqrt{17}}{\sqrt{2} \times 17} \\ \frac{3 \times \sqrt{17}}{\sqrt{2} \times 17} \end{bmatrix}$$

$$q_3 = \begin{bmatrix} -3/\sqrt{34} \\ -4/\sqrt{34} \\ 3/\sqrt{34} \end{bmatrix}$$

$$A = Q \cdot R ; \quad Q = [q_1 \; q_2 \; q_3] \quad R = \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 & 5/\sqrt{34} & -3/\sqrt{34} \\ 3/5 & -12/5\sqrt{34} & -4/\sqrt{34} \\ 4/5 & 9/5\sqrt{34} & 3/\sqrt{34} \end{bmatrix} \qquad A = \begin{array}{c} \begin{bmatrix} 0 & 1 & 1 \\ 3 & 0 & 1 \\ 4 & 1 & 3 \end{bmatrix} \\ \; a_1 \;\; a_2 \;\; a_3 \end{array}$$

$$q_1^T a_1 = \begin{bmatrix} 0 & 3/5 & 4/5 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = 5$$

$$q_1^T \cdot a_2 = \begin{bmatrix} 0 & 3/5 & 4/5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 4/5$$

$$a_1^T a_3 = \begin{bmatrix} 0 & 3/5 & 4/5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = 3$$

$$a_2^T a_2 = \begin{bmatrix} \frac{5}{\sqrt{34}} & \frac{-12}{5\sqrt{34}} & \frac{9}{5\sqrt{34}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \frac{5}{\sqrt{34}} + \frac{9}{5\sqrt{34}}$$

$$\frac{25+9}{5\sqrt{34}} = \frac{34}{5\sqrt{34}} = \frac{\sqrt{34}}{5}$$

$$a_2^T a_3 = \begin{bmatrix} \frac{5}{\sqrt{34}} & \frac{-12}{5\sqrt{34}} & \frac{9}{5\sqrt{34}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \left[\frac{5}{\sqrt{34}} - \frac{12}{5\sqrt{34}} + \frac{27}{5\sqrt{34}}\right]$$

$$= \frac{25-12+27}{5\sqrt{34}} = \frac{40}{5\sqrt{34}} = \frac{8}{\sqrt{34}}$$

$$a_3^T a_3 = \begin{bmatrix} \frac{-3}{\sqrt{34}} & \frac{-4}{\sqrt{34}} & \frac{3}{\sqrt{34}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \frac{-3}{\sqrt{34}} - \frac{4}{\sqrt{34}} + \frac{9}{\sqrt{34}} = \frac{2}{\sqrt{34}}$$

$$\therefore \quad R = \begin{bmatrix} 5 & 4/5 & 3 \\ 0 & \frac{\sqrt{34}}{5} & \frac{8}{\sqrt{34}} \\ 0 & 0 & \frac{2}{\sqrt{34}} \end{bmatrix} \qquad A = Q \cdot R$$

$$\frac{12}{25} - \frac{12}{25}$$

$$\frac{9}{5} - \frac{96}{5\sqrt{34}\sqrt{34}} - \frac{8}{}$$

$$\frac{9\sqrt{34} - 96 - 40}{5\sqrt{34}}$$

$$\underbrace{\begin{bmatrix} 0 & 1 & 1 \\ 3 & 0 & 1 \\ 4 & 1 & 3 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 0 & 5/\sqrt{34} & \frac{-3}{\sqrt{34}} \\ 3/5 & -12/5\sqrt{34} & -4/\sqrt{34} \\ 4/5 & \frac{9}{5\sqrt{34}} & 3/\sqrt{34} \end{bmatrix}}_{Q} \cdot \begin{bmatrix} 5 & 4/5 & 3 \\ 0 & \sqrt{34}/5 & 8/\sqrt{34} \\ 0 & 0 & 2/\sqrt{34} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 \\ 3 & 0 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

34 × 12 = 68, 34 × = 408

306 − 236 = 70

$$\frac{16}{25} + \frac{9}{25}$$

34 × 9 = 306

$$\frac{12}{5} + \frac{72}{170} + \frac{6}{34} = \frac{408+72+30}{170} = \frac{510}{170}$$

$$\frac{40}{34} - \frac{6}{34}$$

$$\frac{9}{5} - \frac{096}{170} - \frac{8}{34} = \frac{306-196-40}{170}$$

⑥

6) Find an orthonormal set q_1, q_2, q_3 for which q_1 and q_2 span the column space of

$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$. What fundamental subspace contains q_3? What is the least squares solution $Ax = b$ if $b = (1, 2, 7)$?

<u>solution:-</u> $a = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ $b = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$

$$q_1 = \frac{a}{\|a\|} = \frac{1}{\sqrt{4+4+1}} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

$$q_1 = \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

$$q_2 = \frac{e_2}{\|e_2\|} \Rightarrow e_2 = b - (q_1^T b) q_1$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \left(\frac{1}{3} - \frac{2}{3} - \frac{8}{3}\right) \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - (-3) \begin{pmatrix} 1/3 \\ 2/3 \\ -2/3 \end{pmatrix}$$

$$e_2 = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}; \|e_2\| = 3$$

$$\therefore q_2 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$$

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Let $a_3 = e_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. To find a_3, let the vector be assumed as orthogonal to the plane spanned by a_1 and a_2 such that $a_1^T \cdot e_3 = 0$ & $a_2^T \cdot e_3 = 0$.

$$\begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \qquad \begin{bmatrix} 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$x + 2y - 2z = 0 \qquad 2x + y + 2z = 0$$

$$\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 6 \end{bmatrix}$$

$$-3y + 6z = 0$$
$$-3y = -6z$$
$$\boxed{y = 2z}$$

$$x + 2y - 2z = 0$$
$$x + 2(2z) - 2z = 0$$
$$x + 4z - 2z = 0$$
$$\boxed{x = -2z}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2z \\ 2z \\ z \end{pmatrix} \Rightarrow Q = \begin{pmatrix} 1/3 & 2/3 & -2z \\ 2/3 & 1/3 & 2z \\ -2/3 & 2/3 & z \end{pmatrix}$$

rows should be orthonormal.

$$\therefore \sqrt{1/9 + 4/9 + 4z^2} = 1$$
$$1/9 + 4/9 + 4z^2 = 1$$
$$4z^2 = 1 - 5/9$$
$$4z^2 = 4/9$$
$$z^2 = 1/9 \quad \boxed{z = 1/3}$$

$$\therefore Q = \begin{pmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ -2/3 & 2/3 & 1/3 \end{pmatrix}$$

(8) consider $Ax = b \Rightarrow A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$

We can verify that $Ax = b$ is not solvable. Hence we consider $R \cdot \hat{x} = Q^T b$ where $R = \begin{bmatrix} q_1^T a & q_1^T b \\ 0 & q_2^T b \end{bmatrix}$

$$q_1^T a = \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = 1/3 + 4/3 + 4/3 = 3$$

$$q_1^T b = \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = 1/3 - 2/3 - 8/3 = -3$$

$$q_2^T b = \begin{bmatrix} 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = 2/3 - 1/3 + 8/3 = 3$$

$$R = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix}$$

$$R\hat{x} = Q^T b$$

$$\begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix} \hat{x} = \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 + 4/3 - 14/3 \\ 2/3 + 2/3 + 14/3 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$\therefore \quad \hat{x} = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix}^{-1} \begin{bmatrix} -3 \\ 6 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -9 + 18 \\ 0 + 18 \end{bmatrix}$$

$$\therefore \quad \hat{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

12) Find the matrix A whose eigen values are 2 and 5 and whose eigen vectors are $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}1\\1\end{pmatrix}$ respectively using $S\Lambda S^{-1}$?

Solution:- $S = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$; $S^{-1} = \frac{1}{1}\begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}$; $\Lambda = \begin{bmatrix}2 & 0\\0 & 5\end{bmatrix}$

$$\therefore A = S\Lambda S^{-1} = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}2 & 0\\0 & 5\end{bmatrix}\begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}2 & 5\\0 & 5\end{bmatrix}\begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}$$

$$\boxed{A = \begin{bmatrix}2 & 3\\0 & 5\end{bmatrix}}$$

14) Find all eigen values & eigen vectors of A given by $A = \begin{pmatrix}1 & 1 & 1\\1 & 1 & 1\\1 & 1 & 1\end{pmatrix}$ and write two different diagonalizing matrices S.

Solution:- $|A-\lambda I| = 0$ $\begin{vmatrix}1-\lambda & 1 & 1\\1 & 1-\lambda & 1\\1 & 1 & 1-\lambda\end{vmatrix} = 0$

$$(1-\lambda)\left[(1-\lambda)^2-1\right] - 1\left[(1-\lambda)-1\right] + 1\left[1-(1-\lambda)\right] = 0$$

$$(1-\lambda)\left[\cancel{1}+\lambda^2-2\lambda-\cancel{1}\right] - 1\left[\cancel{1}-\lambda-\cancel{1}\right] + 1\left[\cancel{1}-\cancel{1}+\lambda\right] = 0$$

$$(1-\lambda)\left[\lambda^2-2\lambda\right] + \lambda + \lambda = 0$$

$$\lambda^2 - \cancel{2\lambda} - \lambda^3 + 2\lambda^2 + \cancel{2\lambda} = 0$$

$$-\lambda^3 + 3\lambda^2 = 0$$

$$\lambda^3 - 3\lambda^2 = 0 \Rightarrow \lambda^2(\lambda-3) = 0 \quad \therefore \lambda = 0, 0, 3$$

(10) consider $[A-\lambda I]X=0$

$$\left.\begin{aligned}(1-\lambda)x+y+z&=0\\ x+(1-\lambda)y+z&=0\\ x+y+(1-\lambda)z&=0\end{aligned}\right\}\text{three equations}$$

case 1) For $\lambda=0$

$$x+y+z=0 \quad ; \quad x=-y-z$$
$$x+y+z=0 \qquad \text{let } y=k_1 \ \& \ z=k_2$$
$$x+y+z=0$$

$\therefore \quad x=-k_1-k_2$

$$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-k_1-k_2\\k_1\\k_2\end{pmatrix}$$

let $k_1=k_2=1$

$$\therefore \quad X_1=\begin{pmatrix}-2\\1\\1\end{pmatrix}$$

case 2) For $\lambda=0$

let $k_1=0 \ \& \ k_2=1$ (Or) let $k_1=1 \ \& \ k_2=0$

$$X_2=\begin{pmatrix}-1\\0\\1\end{pmatrix} \qquad X_2=\begin{pmatrix}-1\\1\\0\end{pmatrix}$$

case 3) For $\lambda=3$

$$\begin{aligned}-2x+y+z&=0\\ x-2y+z&=0\\ x+y-2z&=0\end{aligned} \qquad \begin{bmatrix}-2&1&1\\1&-2&1\\1&1&-2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow[2R_3+R_1]{2R_2+R_1} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix} \xrightarrow{R_3+R_2} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

x, y = pivot variables ; z = free variable

$$-3y+3z=0 \qquad -2x+y+z=0$$

$$-3y=-3z \qquad -2x+z+z=0$$

$$\boxed{y=z} \qquad -2x+2z=0$$

$$-2x+2z=0$$

$$-2x=-2z$$

$$\boxed{x=z}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} z \\ z \\ z \end{pmatrix}$$

let $z=K_1=1$; $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = X_3$

$\therefore$ Two different diagonalizing matrices are:-

$$S_1 = \begin{pmatrix} -2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} \text{ or } S_2 = \begin{pmatrix} -2 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

13) Factor $A = \begin{pmatrix} 3 & 4 \\ -1 & 8 \end{pmatrix}$ into $S \Lambda S^{-1}$ and hence compute A^8?

Solution:-

$$|A-\lambda I|=0$$

$$\begin{vmatrix} 3-\lambda & 4 \\ -1 & 8-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(8-\lambda)+4=0$$

$$24-3\lambda-8\lambda+\lambda^2+4=0$$

$$\lambda^2-11\lambda+28=0$$

$$\lambda^2-4\lambda-7\lambda+28=0$$

$$\lambda(\lambda-4)-7(\lambda-4)=0$$

$$\lambda=4 \text{ or } \lambda=7$$

(12) $[A-\lambda I]x=0$

$\Rightarrow (3-\lambda)x+4y=0$

$-x+(8-\lambda)y=0$

case 1) for $\lambda=4$

$-x+4y=0$

$-x+4y=0$

$-x=-4y$

$\boxed{x=4y}$

let $y=1 \Rightarrow x=4$

$x_1=\begin{pmatrix}4\\1\end{pmatrix}$

case 2) for $\lambda=7$

$-4x+4y=0$

$-x+y=0$

$\boxed{x=y}$

$x_2=\begin{pmatrix}1\\1\end{pmatrix}$

$\therefore \quad S=\begin{pmatrix}4&1\\1&1\end{pmatrix} \qquad S^{-1}=\frac{1}{3}\begin{pmatrix}1&-1\\-1&4\end{pmatrix}$

$\Rightarrow S^{-1}=\begin{pmatrix}1/3&-1/3\\-1/3&4/3\end{pmatrix}$

$A=S\Lambda S^{-1}$

$A=\begin{pmatrix}4&1\\1&1\end{pmatrix}\begin{pmatrix}4&0\\0&7\end{pmatrix}\begin{pmatrix}1/3&-1/3\\-1/3&4/3\end{pmatrix}$

$A^8=\begin{pmatrix}4&1\\1&1\end{pmatrix}\begin{pmatrix}4^8&0\\0&7^8\end{pmatrix}\begin{pmatrix}1/3&-1/3\\-1/3&4/3\end{pmatrix}$

$=\begin{pmatrix}4^8&7^8\\4^8&7^8\end{pmatrix}\begin{pmatrix}1/3&-1/3\\-1/3&4/3\end{pmatrix}=\begin{pmatrix}\frac{4^8-7^8}{3}&-\frac{4^8}{3}+4\cdot7^8/3\\\frac{4^8-7^8}{3}&-\frac{4^8}{3}+4\cdot7^8/3\end{pmatrix}$

9) Find the eigen values and the corresponding eigen vectors of $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{pmatrix}$

Solution for ii) refer lecture notes - page 23

i) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)\{(2-\lambda)^2\} + 1\{-(2-\lambda)\} = 0$$

$$(2-\lambda)\{4+\lambda^2-4\lambda\} + (-2+\lambda) = 0$$

$$8 + 2\lambda^2 - 8\lambda - 4\lambda - \lambda^3 + 4\lambda^2 + \lambda - 2 = 0$$

$$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$(-1)^3 - 6(1) + 11(-1) - 6 \neq 0$$

$$1 - 6 + 11 - 6 = 0$$

$\therefore$ $\lambda = 1$ is one root

$\lambda = 1$	1	-6	11	-6
	0	1	-5	6
	1	-5	6	$\boxed{0}$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda-3) - 2(\lambda-3) = 0$$

$$(\lambda-2)(\lambda-3) = 0$$

$$\lambda = 2 \text{ or } \lambda = 3$$

$\therefore$ eigen values $= \lambda = 1, 2, 3$

(14)

$$[A - \lambda I]x = 0$$

$$(2-\lambda)x + z = 0$$

$$(2-\lambda)y = 0$$

$$x + (2-\lambda)z = 0$$

<u>for $\lambda = 1$</u>

$$x + z = 0$$

$$y = 0$$

$$x + z = 0$$

$$x = -z$$

let $z = 1 \Rightarrow \cancel{y}\ x = -1$

$$\therefore \quad X_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

<u>for $\lambda = 2$</u>

$$z = 0 \qquad X_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$y = 0$$

$$x = 0$$

<u>for $\lambda = 3$</u>

$$-x + z = 0 \qquad X_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$-y = 0$$

$$x - z = 0$$

$$\boxed{x = z}$$

$$\therefore \quad X = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

10) Find the eigen values and eigen vectors of $A = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$. Verify that the trace equals the sum of eigen values and determinant equals their product. If we shift A to $A-7I$ what are the eigen values & eigen vectors and how are they related to A? Also find the eigen values of A^2, A^{-1}?

Solution:- $|A-\lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 4 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 4 = 0$$

$$1 + \lambda^2 - 2\lambda - 4 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda^2 - 3\lambda + \lambda - 3 = 0$$

$$\lambda(\lambda - 3) + 1(\lambda - 3) = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$$\lambda = -1 ; \lambda = 3$$

Consider $[A - \lambda I]X = 0$

$$(1-\lambda)x + 4y = 0$$

$$x + (1-\lambda)y = 0$$

Case 1) for $\lambda = -1$

$$2x + 4y = 0$$

$$x + 2y = 0$$

$$x = -2y$$

let $y = 1$

$\therefore x = -2$

$\therefore X_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Case 2 for $\lambda = 3$

$$-2x + 4y = 0$$

$$x - 2y = 0$$

$$x = 2y$$

let $y = 1$

$\therefore x = 2$

$\therefore X_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Sum of eigen values $= -1 + 3 = 2$

Trace = Sum of principal diagonal elements $= 2$

(16)

Determinant of $A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$

$$|A| = 1-4 = -3$$

Product of eigen values $= -1 \times 3 = -3$

If A is shifted to $A-7I$

$$A-7I = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 4 \\ 1 & -6 \end{bmatrix} = B$$

$$|B-\lambda I| = 0 \Rightarrow \begin{vmatrix} -6-\lambda & 4 \\ +1 & -6-\lambda \end{vmatrix} = 0$$

$(-6-\lambda)^2 = 0$ $\quad 36+\lambda^2+12\lambda = 0$

$\lambda^2+6\lambda+6\lambda+36 = 0$

$\lambda(\lambda+6)+6(\lambda+6) = 0$

$(\lambda+6)(\lambda+6) = 0$

$\lambda = -6$ or $\lambda = -6$

$\begin{vmatrix} -6-\lambda & 4 \\ +1 & -6-\lambda \end{vmatrix} = 0$ $\quad (-6-\lambda)^2 \pm 4 = 0$

$36+\lambda^2+12\lambda \pm 4 = 0$

$\lambda^2+12\lambda+32 = 0$

$\lambda^2-8\lambda-4\lambda+32 = 0$

$\lambda(\lambda-8)-4(\lambda-8) = 0$

$\lambda = 8$ or $\lambda = 4$

$[A-\lambda I]x = 0$

$(-6-\lambda)x+4y = 0$

$x+(-6-\lambda)y = 0$

<u>case 1)</u> $\lambda = 8$

$-14x$

$(-6-\lambda)^2-4 = 0$

$36+\lambda^2+12\lambda-4 = 0$

$\lambda^2+12\lambda+32 = 0$

$\lambda^2+8\lambda+4\lambda+32 = 0$

$\lambda(\lambda+8)+4(\lambda+8) = 0$

$(\lambda+8)(\lambda+4) = 0$

$\lambda = -8 \; ; \; \lambda = -4$

i.e $\lambda = -8$ or $\lambda = -4$

$(-6-\lambda)x + 4y = 0$

$x + (-6-\lambda)y = 0$

case 1) $\lambda = -4$

$-10x + 4y = 0$

$x - 10y = 0$

$$\begin{bmatrix} 1 & -10 \\ -10 & 4 \end{bmatrix} \xrightarrow{R_2 + 10R_1} \begin{bmatrix} 1 & -10 \\ 0 & -96 \end{bmatrix}$$

$x - 10y = 0$

$x = 10y$

$X_1 = \begin{bmatrix} 10 \\ 1 \end{bmatrix}$

case 2) $\lambda = -8$

$-14x + 4y = 0$

$x - 14y = 0$

$X_2 = \begin{bmatrix} 14 \\ 1 \end{bmatrix}$

case 1) $-2x + 4y = 0$

$x - 2y = 0$

$x = 2y$

$X_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

case 2) $2x + 4y = 0$

$x + 2y = 0$

$X_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Eigen values of $A^2 = \lambda^2 = (1, 9)$

Eigen values of $A^{-1} = \frac{1}{\lambda} = (-1, 1/3)$

15) Find the matrices S & S^{-1} to diagonalize $A = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}$ what are limits of Λ^k and $S\Lambda^k S^{-1}$ as $k \to \infty$?

Solution $A = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}$

$|A - \lambda I| = 0 \quad \begin{vmatrix} 0.6-\lambda & 0.4 \\ 0.4 & 0.6-\lambda \end{vmatrix} = 0$

$(0.6-\lambda)^2 - 0.4^2 = 0$

$0.6^2 + \lambda^2 - 2(0.6)(\lambda) - 0.4^2 = 0$

$0.36 + \lambda^2 - 1.2\lambda - 0.16 = 0$

$\lambda^2 - 1.2\lambda + 0.20 = 0$

$$\frac{1.2 \pm \sqrt{1.2^2 - 4(1)(0.20)}}{2} = \frac{1.2 \pm 0.8}{2}$$

eigen values are 1 & 0.2

(18) $[A-\lambda I]X = 0$

$(0.6-\lambda)x + 0.4y = 0$

$0.4x + (0.6-\lambda)y = 0$

<u>case 1)</u> $\lambda = 0.2$

$0.4x + 0.4y = 0$

$0.4x + 0.4y = 0$

$0.4x = -0.4y$

let $y = 1$

$x = -1$

$\therefore X_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

<u>case 2)</u> $\lambda = 1$

$-0.4x + 0.4y = 0$

$0.4x - 0.4y = 0$

$x = y$

let $y = 1$

$\Rightarrow x = 1$

$\therefore X_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\therefore S = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ $S^{-1} = \frac{1}{-2}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

$$A = S\Lambda S^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0.2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$A = \begin{pmatrix} -0.2 & 1 \\ 0.2 & 1 \end{pmatrix}\begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$A = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$$

$$A^K = S\Lambda^K S^{-1} = S\begin{pmatrix} 0.2 & 0 \\ 0 & 1 \end{pmatrix}^K \cdot S^{-1}$$

11) Find the characteristic equation and hence find the inverse of $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ using the cayley hamilton theorem:

Solution :- characteristic equation :-

$$|A-\lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 1 \\ 2 & 1-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$\cancel{(1-\lambda)\{2(2-\lambda)-1\}-2}$$

$$(1-\lambda)\{(1-\lambda)(2-\lambda)-1\}-2\{2(2-\lambda)-1\}+1\{2-(1-\lambda)\} = 0$$

$$(1-\lambda)\{2-\lambda-2\lambda+\lambda^2-1\}-2\{4-2\lambda-1\}+1\{2-1+\lambda\} = 0$$

$$(1-\lambda)\{\lambda^2-3\lambda+1\}-2\{3-2\lambda\}+1\{1+\lambda\} = 0$$

$$\lambda^2-3\lambda+1-\lambda^3+3\lambda^2-\cancel{\lambda}-6+4\lambda+1+\cancel{\lambda} = 0$$

$$-\lambda^3+4\lambda^2+\lambda-4 = 0$$

$$\lambda^3-4\lambda^2-\lambda+4 = 0$$

According to cayley Hamilton theorem :-

λ to be replaced by A :-

$$A^3-4A^2-A+4I = 0 \quad -(1)$$

multiplying (1) by A^{-1}

$$A^2-4A-A\cdot A^{-1}+4IA^{-1} = 0$$

$$4A^{-1} = -A^2+4A+I$$

$$A^{-1} = \frac{1}{4}[-A^2+4A+I]$$

$$A^{-1} = \frac{1}{4}\left[-\left\{\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}\right\}+4\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}+\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right]$$

(20) $$A^2 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} * \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix}$$

$$A^{-1} = \frac{1}{4}\left[-\begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix} + 4\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right]$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -6 & -5 & -5 \\ -5 & -6 & -5 \\ -5 & -5 & -6 \end{bmatrix} + \begin{pmatrix} 4 & 8 & 4 \\ 8 & 4 & 4 \\ 4 & 4 & 8 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -1 & 3 & -1 \\ 3 & -1 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

1) Verify that $(-2,1,3,0)$ $(0,-3,1,-6)$ $(-2,-4,0,2)$ is an orthogonal set of vectors in $\mathbb{R}^4$ and use it to construct an orthonormal set of vectors in $\mathbb{R}^4$.

<u>solution</u>

$$A = \begin{array}{c} \begin{pmatrix} -2 & 0 & -2 \\ 1 & -3 & -4 \\ 3 & 1 & 0 \\ 0 & -6 & 2 \end{pmatrix} \\ \quad a \quad\; b \quad\; c \end{array}$$

for these vectors to be orthogonal, $a^Tb = 0$, $a^Tc = 0$ & $b^Tc = 0$

$$\begin{bmatrix} -2 & 1 & 3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -3 \\ 1 \\ -6 \end{bmatrix} = -3+3 = 0 \Rightarrow a^Tb = 0$$

$$\begin{bmatrix} -2 & 1 & 3 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ -4 \\ 0 \\ 2 \end{bmatrix} = 4-4 = 0 \Rightarrow a^Tc = 0$$

$$[0\ -3\ 1\ -6]\begin{bmatrix}-2\\-4\\0\\2\end{bmatrix} = 12-12 = 0$$

$\therefore$ a, b, c are orthogonal vectors in $\mathbb{R}^4$.

<u>orthonormal set :-</u>

$$\alpha_1 = \frac{1}{\sqrt{14}}\begin{pmatrix}-2\\1\\3\\0\end{pmatrix};\quad \alpha_2 = \frac{1}{\sqrt{46}}\begin{pmatrix}0\\-3\\1\\-6\end{pmatrix};\quad \alpha_3 = \frac{1}{2\sqrt{6}}\begin{bmatrix}-2\\-4\\0\\2\end{bmatrix}$$

2) Let $W = \{(a,b,b) / a,b$ are real$\}$ and let $v = (3,2,6)$.

i) Find an orthonormal basis for W

ii) Find the Projection of v onto W, say V1

iii) Decompose v into a sum of two vectors V1+V2 where V2 is projection of v onto $W^\perp$?

<u>solution :-</u> $W = \begin{pmatrix}a\\b\\b\end{pmatrix} = a\begin{pmatrix}1\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\1\\1\end{pmatrix}$

i) orthonormal basis for W = $\begin{pmatrix}1\\0\\0\end{pmatrix}$ & $\begin{pmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$

$$Q = \begin{bmatrix}1 & 0\\0 & 1/\sqrt{2}\\0 & 1/\sqrt{2}\end{bmatrix}$$

ii) Projection of v onto W is given by :-

$$v_1 = P.v = Q.Q^T.v \quad \begin{bmatrix}1 & 0\\0 & 1/\sqrt{2}\\0 & 1/\sqrt{2}\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix} * \begin{bmatrix}3\\2\\6\end{bmatrix}$$

$$v_1 = \begin{bmatrix}1 & 0 & 0\\0 & 1/2 & 1/2\\0 & 1/2 & 1/2\end{bmatrix}\begin{bmatrix}3\\2\\6\end{bmatrix} = \begin{bmatrix}3\\4\\4\end{bmatrix}$$

iii) $v_2 = v - v_1 = \begin{pmatrix}3\\2\\6\end{pmatrix} - \begin{pmatrix}3\\4\\4\end{pmatrix} = \begin{pmatrix}0\\-2\\2\end{pmatrix}$

(22)

4) If W is a subspace spanned by the orthogonal vectors $(2,5,-1)$ and $(-2,1,1)$ find the point in W that is closest to $(1,2,3)$?

Solution $\quad \omega_1 = (2,5,-1)$; $\omega_2 = (-2,1,1)$; $b = (1,2,3)$

$$P = \frac{b^T \cdot \omega_1}{\omega_1^T \cdot \omega_1} \cdot \omega_1 + \frac{b^T \cdot \omega_2}{\omega_2^T \cdot \omega_2} \cdot \omega_2$$

$$P = \frac{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}}{30} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}}{6} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$P = \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \frac{3}{6}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$P = \frac{3}{10}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 3/5 \\ 3/2 \\ -3/10 \end{bmatrix} + \begin{bmatrix} -1 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 3/5 - \frac{1}{1} \\ 3/2 + 1/2 \\ \frac{-3}{10} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3-5}{5} \\ 2 \\ \frac{-3+5}{10} \end{bmatrix}$$

$$P = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

5) What multiple of $a_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ should be subtracted from $a_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$ to make the result orthogonal to a_1? Factor $A = QR$ with orthonormal vectors in Q?

Solution :- $a_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$; $a_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$; $q_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$$v_2 = a_2 - \frac{a_1^T \cdot a_2}{a_1^T a_1} \cdot a_1$$

$$v_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix} - \frac{\begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} - \frac{4}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \end{pmatrix} \Rightarrow v_2 = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$q_2 = \frac{1}{2\sqrt{2}}\begin{pmatrix} 2 \\ -2 \end{pmatrix} \Rightarrow q_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$\therefore Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \qquad R = \begin{bmatrix} q_1^T a_1 & q_1^T a_2 \\ 0 & q_2^T a_2 \end{bmatrix}$$

$$q_1^T \cdot a_1 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sqrt{2}$$

$$q_1^T \cdot a_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} 4 \\ 0 \end{bmatrix} = 2\sqrt{2}$$

$$q_2^T a_2 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}\begin{pmatrix} 4 \\ 0 \end{pmatrix} = 2\sqrt{2}$$

$$\therefore A = QR = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}\begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix} = \begin{pmatrix} 1 & 4 \\ 1 & 0 \end{pmatrix}$$

(24)

8) use Gram-Schmidt orthogonalization process to transfer the basis $\begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix}\begin{pmatrix}0&1\\0&1\end{pmatrix}\begin{pmatrix}1&0\\0&1\end{pmatrix}$ of $M_{2\times2}$ in to an orthogonal basis if $A^TB = tr(AB^T)$?

Solution :-

$$B_1 = A_1 = \begin{pmatrix}1&1\\0&0\end{pmatrix}$$

$$B_2 = A_2 - \frac{\langle A_2, B_1\rangle}{\langle B_1, B_1\rangle} \cdot B_1$$

$$\langle A_2, B_1\rangle = A_2^T \cdot B_1 = \begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}1&1\\0&0\end{pmatrix}$$

$$B_1^T \cdot B_1 = \begin{pmatrix}1&0\\1&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}1&1\\1&1\end{pmatrix}$$

$$B_2 = \begin{pmatrix}1&0\\0&0\end{pmatrix} - \begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix}$$

$$B_2 =$$

$$A_2 \cdot B_1^T = \begin{pmatrix}1&0\\1&0\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix} = \begin{pmatrix}1&0\\1&0\end{pmatrix}$$

$A^TB = tr(AB^T)$. $\therefore$ $\langle A_2, B_1\rangle = 1$

$$\langle B_1, B_1\rangle = tr(B_1 \cdot B_1^T)$$

$$B_1 \cdot B_1^T = \begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix} = \begin{pmatrix}2&0\\0&0\end{pmatrix} ; Tr(B_1 \cdot B_1^T) = 2$$

$$\therefore B_2 = \begin{pmatrix}1&0\\1&0\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}1/2&-1/2\\1&0\end{pmatrix}$$

$$B_3 = A_3 - \frac{\langle A_3, B_1\rangle}{\langle B_1, B_1\rangle} \cdot B_1 - \frac{\langle A_3, B_2\rangle}{\langle B_2, B_2\rangle} \cdot B_2$$

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$$B_3 = \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix} - \frac{\langle A_3, B_1\rangle}{\langle B_1, B_1\rangle}\begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} - \frac{\langle A_3, B_2\rangle}{\langle B_2, B_2\rangle}\begin{pmatrix}1/2 & -1/2\\ 1 & 0\end{pmatrix}$$

$$\langle A_3, B_1\rangle = A_3^T \cdot B_1 = Tr\left(A_3 \cdot B_1^T\right)$$

$$A_3 \cdot B_1^T = \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix} = \begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix}$$

$$Tr\left(A_3 \cdot B_1^T\right) = 1$$

$$\langle B_1, B_1\rangle = 2$$

$$\langle A_3, B_2\rangle = A_3^T \cdot B_2 \overset{Tr}{=} \left\{A_3 \cdot B_2^T\right\} \Rightarrow$$

$$A_3 \cdot B_2^T = \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1/2 & 1\\ -1/2 & 0\end{pmatrix} = \begin{pmatrix}-1/2 & 0\\ -1/2 & 0\end{pmatrix}$$

$$Tr\left\{A_3 \cdot B_2^T\right\} = -1/2$$

$$\langle B_2, B_2\rangle = B_2^T \cdot B_2 = Tr\left\{B_2 \cdot B_2^T\right\}$$

$$B_2 \cdot B_2^T = \begin{pmatrix}1/2 & -1/2\\ 1 & 0\end{pmatrix}\begin{pmatrix}1/2 & 1\\ -1/2 & 0\end{pmatrix} = \begin{pmatrix}1/2 & 1/2\\ 1/2 & 1\end{pmatrix}$$

$$Tr\left\{B_2 \cdot B_2^T\right\} = 3/2$$

$$\therefore \quad B_3 = \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} - \frac{(-1/2)}{(3/2)}\begin{pmatrix}1/2 & -1/2\\ 1 & 0\end{pmatrix}$$

$$\frac{1}{2} - \frac{1}{6} = \frac{3-1}{6}$$

$$B_3 = \begin{pmatrix}0 & 1\\ 0 & 1\end{pmatrix} - \begin{pmatrix}1/2 & 1/2\\ 0 & 0\end{pmatrix} + \begin{pmatrix}1/6 & -1/6\\ 1/3 & 0\end{pmatrix}$$

$$-\frac{1}{2} + \frac{1}{6} = \frac{-3+1}{6} = \frac{-1}{3}$$

$$B_3 = \begin{pmatrix}-1/3 & 1/3\\ 1/3 & 1\end{pmatrix}$$

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$$B_4 = A_4 - \frac{\langle A_4, B_1\rangle}{\langle B_1, B_1\rangle}\cdot B_1 - \frac{\langle A_4, B_2\rangle}{\langle B_2, B_2\rangle}\cdot B_2 - \frac{\langle A_4, B_3\rangle}{\langle B_3, B_3\rangle}\cdot B_3$$

$$\frac{1}{9}+\frac{1}{1} = \frac{1+9}{9}$$

$$\frac{-1}{9}+\frac{1}{3} = \frac{-1+3}{9}$$

$\boxed{\langle B_1, B_1\rangle = 2}$; $\boxed{\langle B_2, B_2\rangle = 3/2}$; $\langle B_3, B_3\rangle = B_3^T\cdot B_3$

$$B_3^T\cdot B_3 = Tr\{B_3\cdot B_3^T\} ; \quad B_3\cdot B_3^T = \begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix}\begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix} = \begin{pmatrix} 2/9 & 2/9 \\ 2/9 & 10/9\end{pmatrix}$$

$$\boxed{\therefore \langle B_3, B_3\rangle = Tr\{B_3\cdot B_3^T\} = 12/9 = 4/3}$$

$$\langle A_4, B_1\rangle = A_4^T\cdot B_1 = Tr\{A_4\cdot B_1^T\}$$

$$A_4\cdot B_1^T = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 0\end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0\end{pmatrix} ; \quad Tr\{A_4\cdot B_1^T\} = 1$$

$$\boxed{\therefore \langle A_4, B_1\rangle = 1}$$

$$\langle A_4, B_2\rangle = A_4^T\cdot B_2 = Tr\{A_4\cdot B_2^T\}$$

$$A_4\cdot B_2^T = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix}\begin{pmatrix} 1/2 & 1 \\ -1/2 & 0\end{pmatrix} = \begin{pmatrix} 1/2 & 1 \\ -1/2 & 0\end{pmatrix} ; \quad Tr\{A_4\cdot B_2^T\} = 1/2$$

$$\boxed{\therefore \langle A_4, B_2\rangle = 1/2}$$

$$\langle A_4, B_3\rangle = A_4^T\cdot B_3 = Tr\{A_4\cdot B_3^T\}$$

$$A_4\cdot B_3^T = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix}\begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix} = \begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix} ; \quad Tr\{A_4\cdot B_3^T\}$$

$$= \frac{-1}{3}+\frac{1}{1}$$

$$= \frac{-1+3}{3} = 2/3$$

$$\boxed{\therefore \langle A_4, B_3\rangle = 2/3}$$

$$\therefore B_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 0 & 0\end{pmatrix} - \frac{(1/2)}{(3/2)}\begin{pmatrix} 1/2 & -1/2 \\ 1 & 0\end{pmatrix} - \frac{(2/3)}{(4/3)}\begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix}$$

$$B_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix} - \begin{pmatrix} 1/2 & 1/2 \\ 0 & 0\end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1/2 & -1/2 \\ 1 & 0\end{pmatrix} - \frac{1}{2}\begin{pmatrix} -1/3 & 1/3 \\ 1/3 & 1\end{pmatrix}$$

$$B_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1\end{pmatrix} - \begin{pmatrix} 1/2 & 1/2 \\ 0 & 0\end{pmatrix} - \begin{pmatrix} 1/6 & -1/6 \\ 1/3 & 0\end{pmatrix} - \begin{pmatrix} -1/6 & 1/6 \\ 1/6 & 1/2\end{pmatrix}$$